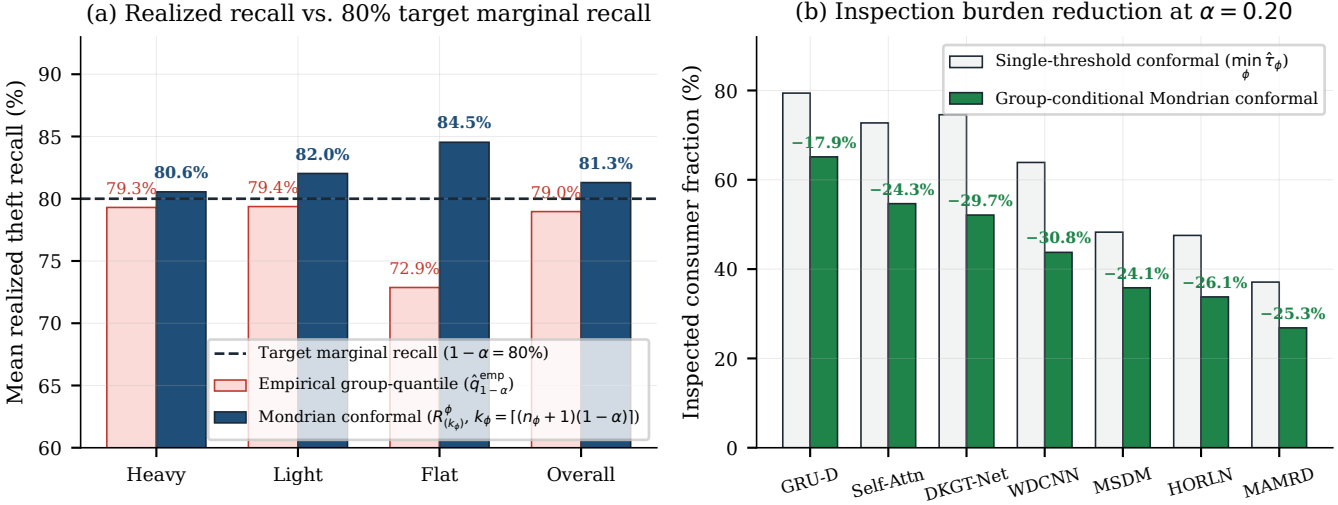
(a) Realized recall vs. 80% target marginal recall
Mean realized theft recall (%)
90
85
80
75
70
65
60
79.3%
80.6%
79.4%
82.0%
72.9%
84.5%
79.0%
81.3%
Target marginal recall ($1-\alpha = 80\%$)
Empirical group-quantile ($\hat{q}^{\text{emp}}_{1-\alpha}$)
Mondrian conformal ($R^{\phi}_{(k_\phi)}$, $k_\phi = \lceil (n_\phi+1)(1-\alpha) \rceil$)
Heavy
Light
Flat
Overall
(b) Inspection burden reduction at $\alpha = 0.20$
Inspected consumer fraction (%)
80
60
40
20
0
Single-threshold conformal ($\min_\phi \hat{\tau}_\phi$)
Group-conditional Mondrian conformal
−17.9%
−24.3%
−29.7%
−30.8%
−24.1%
−26.1%
−25.3%
GRU-D
Self-Attn
DKGT-Net
WDCNN
MSDM
HORLN
MAMRD